

Beyond Zero-Touch: Grand Challenges for Network and Service Management in the Next Decade

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Abstract—The continuous evolution of cloud-native infrastructures, software-defined networking, edge computing, network softwarization, and artificial intelligence is reshaping the landscape of network and service management. While the last decade has witnessed remarkable advances toward autonomous operation through intent-based networking, closed-loop automation, and zero-touch management, current solutions remain limited by fragmented data sources, insufficient reasoning capabilities, scalability constraints, and the lack of trustworthy decision-making mechanisms. These limitations become even more evident as future communication infrastructures evolve toward highly distributed, heterogeneous, and AI-native ecosystems. This paper discusses the major challenges that are expected to shape the next decade of network and service management research. We first review the current state of the art and identify the primary technological bottlenecks that hinder the realization of fully autonomous network operations. We then discuss emerging paradigms—including Agentic Artificial Intelligence, Large Language Models, Digital Twins, semantic telemetry, and intent-driven orchestration—and analyze how they may transform operational workflows. Finally, we outline a research roadmap that highlights open problems whose solutions will likely require entirely new architectural principles rather than incremental improvements over existing technologies.

Index Terms—Network and Service Management, Autonomous Networks, Zero-Touch Management, Agentic AI, Digital Twins, Intent-Based Networking, Network Automation.

I. INTRODUCTION

The management of communication networks has experienced a profound transformation over the last two decades. Traditional network management systems (NMSs), largely based on the Simple Network Management Protocol (SNMP), command-line interfaces (CLI), and manually configured operational policies, have progressively evolved toward programmable, virtualized, and cloud-native infrastructures. The introduction of Software-Defined Networking (SDN) and Network Function Virtualization (NFV) fundamentally changed the networking landscape by decoupling control from forwarding and by replacing dedicated appliances with software-based network functions, enabling unprecedented flexibility and service agility [1]–[3]. More recently, containerization and cloud-native orchestration have further accelerated this evolution, allowing communication services to be deployed dynamically across distributed cloud and edge infrastructures [4].

Despite these technological advances, the operational complexity of modern communication infrastructures has increased dramatically. Contemporary networks integrate heterogeneous

technologies, including multi-cloud platforms, edge computing, programmable data planes, private 5G/6G deployments, Internet of Things (IoT) devices, and non-terrestrial communication systems. Each domain generates large volumes of operational data while exposing distinct management interfaces, telemetry models, and orchestration mechanisms. Consequently, conventional management approaches based on static policies and human-driven operation have become increasingly inefficient, motivating the transition toward intelligent and autonomous management architectures [5], [6].

Automation has therefore emerged as one of the primary research directions in network and service management. Frameworks such as Intent-Based Networking (IBN), Artificial Intelligence for IT Operations (AIOps), and the ETSI Zero-touch Network and Service Management (ZSM) architecture seek to reduce operational complexity by automating configuration, monitoring, optimization, and fault recovery processes. Similarly, ETSI Experiential Networked Intelligence (ENI) extends this vision by incorporating artificial intelligence into management loops, enabling systems to continuously learn from operational experience and improve decision-making over time [7]–[9]. Although these initiatives have significantly reduced manual intervention, current implementations remain predominantly workflow-oriented and still require substantial human supervision during policy definition, model training, and operational validation.

The rapid emergence of Artificial Intelligence, particularly Large Language Models (LLMs), has recently opened new opportunities for network management. General-purpose foundation models have demonstrated remarkable capabilities in natural language understanding, code generation, knowledge retrieval, troubleshooting assistance, and operational documentation. In networking, these capabilities have been explored to translate high-level intents into configurations, explain network anomalies, summarize telemetry information, and assist operators during incident response. At the same time, machine learning techniques continue to improve traffic classification, anomaly detection, resource allocation, and predictive maintenance [10]–[12]. Nevertheless, current AI solutions are still largely designed as isolated decision-support tools rather than autonomous operational entities capable of coordinating complex management tasks across heterogeneous infrastructures.

A particularly promising evolution of this paradigm is represented by **Agentic Artificial Intelligence**, which extends conventional language models with reasoning, planning,

memory, tool invocation, and autonomous execution capabilities. Rather than responding to isolated prompts, autonomous agents are expected to continuously observe network conditions, formulate operational objectives, invoke specialized tools, evaluate alternative strategies, and validate the consequences of their actions before modifying the production environment. This vision naturally complements existing concepts such as Knowledge-Defined Networking (KDN), intent-based management, and autonomic networking by introducing adaptive decision-making mechanisms capable of continuously improving their operational behavior through interaction with dynamic environments [5], [7], [13].

Although recent advances indicate that autonomous network management is becoming increasingly feasible, several fundamental scientific and engineering challenges remain unresolved. Current management platforms still suffer from fragmented observability, limited semantic understanding of operational data, insufficient interoperability across administrative domains, lack of trustworthy AI mechanisms, and restricted support for collaborative decision-making. Furthermore, emerging paradigms such as Digital Twins, semantic telemetry, domain-specific foundation models, and AI-native orchestration are only beginning to mature [14]–[16]. This paper argues that addressing these limitations will require not only the evolution of existing technologies but also the development of entirely new architectural principles capable of supporting trustworthy, explainable, and self-evolving autonomous network and service management over the next decade.

II. CURRENT STATE OF NETWORK AND SERVICE MANAGEMENT

The current landscape of network and service management reflects more than a decade of continuous technological evolution driven by network softwarization, cloud computing, virtualization, and artificial intelligence. Communication infrastructures have progressively shifted from proprietary hardware-based deployments to highly programmable, software-centric ecosystems composed of virtualized network functions, cloud-native services, programmable forwarding devices, and distributed edge platforms. This transformation has significantly improved service agility and deployment flexibility while simultaneously increasing operational complexity. As a result, management platforms have evolved from simple monitoring systems into sophisticated orchestration frameworks responsible for coordinating thousands of interconnected software components operating across multiple administrative domains [1]–[3].

One of the primary enablers of this transformation has been the widespread adoption of cloud-native networking. Containerized Network Functions (CNFs), Kubernetes orchestration, microservice-based Service-Based Architectures (SBAs), and elastic resource provisioning have become the de facto implementation model for modern 5G Core networks and are expected to remain central to future 6G infrastructures. Compared to traditional Virtual Network Functions (VNFs), cloud-

native deployments provide improved scalability, faster service instantiation, and simplified lifecycle management. However, these advantages also increase operational complexity by introducing a much larger number of distributed software components that must be continuously monitored, orchestrated, and optimized. Consequently, service management has become increasingly dependent on intelligent orchestration platforms capable of dynamically adapting to changing workloads while preserving stringent Service-Level Agreements (SLAs) [4], [17].

Programmability has also fundamentally transformed network operation. Software-Defined Networking introduced logically centralized control, enabling dynamic traffic engineering and network reconfiguration through standardized programming interfaces, while Network Function Virtualization decoupled network functions from proprietary appliances, facilitating flexible service deployment and resource allocation [1]–[3]. More recently, programmable data planes based on languages such as P4 have extended programmability into the forwarding plane itself, allowing switches to execute advanced packet processing, telemetry collection, and lightweight in-network intelligence directly at line rate [18]. Despite these advances, most operational decisions remain reactive, relying on monitoring systems that detect performance degradation after service quality has already been affected.

Artificial intelligence has become another fundamental pillar of modern network management. Machine learning algorithms have demonstrated significant benefits for traffic classification, anomaly detection, fault prediction, intrusion detection, resource allocation, and network optimization. Surveys published in recent years consistently show that AI techniques substantially improve operational efficiency while reducing manual intervention [10]–[12]. More recently, Large Language Models have introduced new capabilities for translating intents into network configurations, assisting operators during troubleshooting, generating management scripts, and synthesizing operational knowledge. Nevertheless, these systems largely function as intelligent assistants rather than autonomous management entities. They typically operate independently from one another, possess limited contextual awareness, and lack persistent reasoning mechanisms capable of coordinating long-term operational strategies across heterogeneous infrastructures.

Standardization efforts have evolved in parallel with these technological advances. The ETSI Zero-touch Network and Service Management (ZSM) initiative defines an architectural framework for fully automated service lifecycle management, while ETSI Experiential Networked Intelligence (ENI) introduces AI-driven cognitive functions into operational loops. Similarly, the IETF Autonomic Networking architecture proposes self-managing systems capable of minimizing human intervention through distributed autonomic functions, and TM Forum’s Autonomous Networks initiative provides practical guidance for implementing increasing levels of operational autonomy. In radio access networks, the O-RAN Alliance has introduced intelligent controllers that enable near-real-

Figure 1 – Evolution of Network and Service Management Paradigms

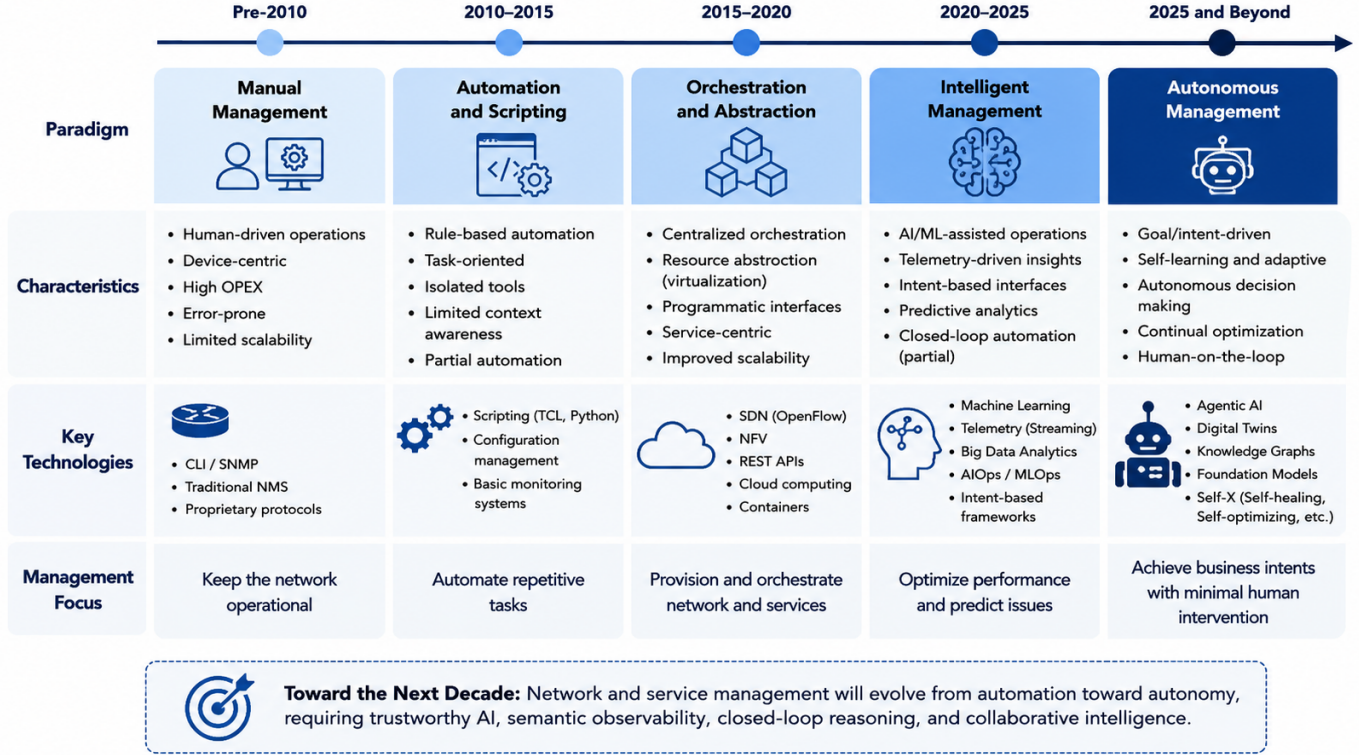


Fig. 1. Current network and service management ecosystem integrating cloud-native infrastructures, AI, orchestration, programmable networks, and observability platforms.

time optimization through AI-driven control loops [5], [6], [8], [9], [16], [19]. Collectively, these initiatives establish the architectural foundations for autonomous operation, although current implementations still rely heavily on manually defined policies, static workflows, and centralized decision logic.

Despite the remarkable progress achieved during the past decade, the current state of network and service management remains transitional. Existing platforms successfully automate repetitive operational procedures but still struggle to understand the semantic meaning of operational data, coordinate decisions across multiple administrative domains, guarantee trustworthy AI behavior, and continuously adapt to previously unseen situations. Moreover, the rapid increase in telemetry generated by cloud-native applications, programmable networks, and edge infrastructures challenges the scalability of existing observability platforms, motivating the adoption of adaptive telemetry frameworks and semantic data representations [14]. These limitations suggest that future management systems must evolve beyond workflow automation toward architectures capable of autonomous reasoning, collaborative intelligence, and continuous knowledge acquisition. Rather than simply extending current technologies, the next generation of management platforms will likely integrate Agentic AI, Digital Twins, semantic telemetry, and networking-specific foundation models into cohesive AI-native operational ecosystems.

TABLE I
CURRENT TECHNOLOGIES AND OPEN LIMITATIONS

Technology	Remaining Challenges
Cloud-native orchestration	Cross-domain coordination, scalability
Network slicing	End-to-end automation and SLA assurance
Programmable data planes	Limited memory and computational resources
Machine Learning	Generalization, explainability, robustness
Large Language Models	Hallucinations, trustworthiness, long-term reasoning
Zero-touch management	Dependence on predefined workflows
Digital Twins	Synchronization fidelity and computational cost
Intent-based networking	Intent ambiguity and conflict resolution

III. CHALLENGES FOR THE NEXT DECADE

The next decade is expected to represent a fundamental transition from automated to autonomous network and service management. While recent technological advances have significantly reduced manual intervention through software-defined infrastructures, virtualization, cloud-native orchestration, and AI-assisted analytics, future communication ecosystems will

operate at a scale and complexity that exceed the capabilities of existing management frameworks. Emerging infrastructures will simultaneously integrate terrestrial and non-terrestrial networks, edge-cloud continuums, AI-native services, Internet of Things (IoT) ecosystems, programmable data planes, and future 6G architectures, generating highly dynamic environments that require continuous adaptation. Under these conditions, traditional closed-loop automation based on predefined workflows will no longer be sufficient, demanding management platforms capable of autonomous reasoning, long-term planning, and collaborative decision-making [5], [6], [8].

One of the most significant challenges concerns the scalability of observability. Modern communication infrastructures continuously generate enormous volumes of telemetry through cloud-native monitoring systems, streaming analytics, distributed logs, traces, and programmable in-band telemetry. Although recent frameworks have considerably improved network visibility, indiscriminate telemetry collection is becoming computationally unsustainable as infrastructures continue to grow in size and heterogeneity. Future management systems must evolve from metric-oriented monitoring toward semantic observability, where telemetry is collected dynamically according to operational objectives, service context, and the expected value of information. Such adaptive observability frameworks should intelligently balance measurement accuracy, communication overhead, storage requirements, and inference latency while preserving sufficient information for autonomous decision-making [14], [18].

Another major challenge lies in achieving trustworthy autonomous intelligence. Existing machine learning solutions have demonstrated impressive performance in traffic classification, anomaly detection, resource allocation, and predictive maintenance; however, most models remain black-box predictors that provide limited interpretability and uncertain behavior under previously unseen operating conditions [10]–[12]. As AI systems gradually transition from advisory tools to autonomous operational entities, management decisions will directly affect network availability, security, and service reliability. Consequently, future autonomous managers must incorporate explainability, uncertainty estimation, causal reasoning, and formal verification into their decision-making processes. Autonomous systems should not only recommend actions but also justify their decisions, evaluate potential risks before execution, and continuously validate whether their actions satisfy operational policies and service-level objectives.

Interoperability represents another critical research challenge. Future communication systems will span multiple cloud providers, private and public 5G/6G infrastructures, programmable transport networks, industrial IoT deployments, satellite communications, and edge computing platforms. Each administrative domain exposes distinct telemetry models, orchestration interfaces, policy languages, and security mechanisms. Although initiatives such as ETSI ZSM, TM Forum Autonomous Networks, and the O-RAN Alliance have established important architectural foundations for management interoperability, current implementations remain largely

domain-specific and rely on manually engineered integration mechanisms [8], [16], [19]. Achieving seamless end-to-end autonomous management across heterogeneous infrastructures will likely require semantic knowledge representation, standardized ontologies, knowledge graphs, and distributed reasoning frameworks capable of coordinating multiple autonomous agents while preserving privacy, security, and administrative autonomy.

Cybersecurity will also become increasingly intertwined with network management. The incorporation of AI into operational workflows introduces entirely new attack surfaces, including telemetry manipulation, data poisoning, adversarial machine learning, prompt injection, model exploitation, and malicious orchestration commands. Traditional security mechanisms designed to protect communication infrastructures are insufficient when management decisions themselves become AI-driven. Therefore, future autonomous networks must integrate security as an intrinsic architectural property rather than as an independent management application. This evolution will require secure telemetry pipelines, trusted execution environments for AI models, authenticated knowledge repositories, continuous policy verification, and resilient multi-agent coordination capable of detecting and mitigating attacks targeting autonomous management processes [5], [12].

Finally, perhaps the greatest challenge is that many of the technologies required to achieve fully autonomous network management have not yet been developed. Current research largely focuses on extending existing paradigms—including larger language models, improved orchestration frameworks, and more accurate machine learning algorithms—but these incremental improvements alone are unlikely to address the fundamental limitations of future communication infrastructures. The next decade will likely witness the emergence of entirely new technologies, including Large Network Models (LNMs) specialized in networking knowledge, continuously synchronized Network Digital Twins, semantic telemetry frameworks, distributed communities of autonomous AI agents, and self-verifying management architectures capable of reasoning over complex operational knowledge [13], [15]. Rather than representing incremental improvements over today’s solutions, these emerging paradigms may redefine the architectural foundations of network and service management, enabling communication infrastructures to evolve toward truly cognitive, trustworthy, and self-adaptive systems.

IV. FINAL REMARKS

The evolution of network and service management over the last two decades has been driven by a continuous effort to reduce operational complexity while improving service reliability, scalability, and resource efficiency. Technologies such as Software-Defined Networking, Network Function Virtualization, cloud-native infrastructures, programmable data planes, and Artificial Intelligence have fundamentally transformed the way communication networks are deployed and operated. Nevertheless, as discussed throughout this paper, these advances have primarily improved automation rather than achieving

genuine autonomy. Existing management platforms remain highly dependent on predefined workflows, manually engineered policies, fragmented operational knowledge, and continuous human supervision. Consequently, despite remarkable technological progress, fully autonomous network operation remains an open scientific challenge.

This paper argued that the next decade will require a fundamental shift from workflow-oriented automation toward reasoning-driven autonomous management. Such a transition will not be achieved merely by increasing the scale of existing AI models or by introducing more sophisticated orchestration mechanisms. Instead, future management systems must integrate reasoning, planning, memory, verification, collaboration, and continuous learning into unified operational architectures capable of understanding high-level service objectives and adapting dynamically to uncertain environments. In this context, autonomy should no longer be interpreted as the automatic execution of predefined actions, but rather as the capability of management systems to formulate, evaluate, and safely execute operational strategies while continuously validating their outcomes.

The analysis presented throughout this work also suggests that several of today’s most promising technologies—including Large Language Models, Digital Twins, intent-based management, semantic observability, and programmable networks—should be regarded as enabling technologies rather than final solutions. Their true potential will emerge only when they become components of larger AI-native ecosystems capable of cooperating through standardized knowledge representations and autonomous decision-making mechanisms. In particular, Agentic Artificial Intelligence appears as one of the most promising paradigms for coordinating distributed operational intelligence across heterogeneous infrastructures. Similarly, Digital Twins are expected to evolve from simulation platforms into continuously synchronized cognitive models supporting predictive operation and autonomous validation. These technologies will likely redefine the operational lifecycle of future communication systems by moving intelligence from isolated management applications into the network itself.

Equally important is the recognition that many of the challenges discussed in this paper remain largely unexplored by current research. Future communication infrastructures will integrate technologies that are still in their infancy, including 6G-native services, integrated terrestrial and non-terrestrial networks, large-scale edge-cloud continuums, quantum communication systems, programmable optical infrastructures, and increasingly autonomous cyber-physical environments. As these technologies mature, entirely new management problems will inevitably emerge, demanding architectural principles that cannot yet be fully anticipated. Therefore, one of the central messages of this paper is that the next decade should not be viewed merely as an extension of current research trends but rather as an opportunity to establish the scientific foundations of a new generation of intelligent communication infrastructures.

Another important observation concerns the increasing role of interdisciplinary research. Future network management will no longer be exclusively a networking problem. Instead, it will require advances spanning distributed systems, artificial intelligence, knowledge representation, software engineering, cybersecurity, formal verification, human-computer interaction, operations research, and even cognitive science. Collaboration among these research communities will be essential to develop trustworthy autonomous systems capable of balancing performance, resilience, security, sustainability, and explainability. Likewise, standardization bodies, industry, and academia must work together to define interoperable architectures that facilitate the safe deployment of autonomous management technologies across heterogeneous operational environments. Such collaboration will be instrumental in transforming isolated research advances into practical solutions capable of supporting future communication ecosystems.

Finally, we believe that the ultimate vision of autonomous network and service management extends beyond achieving fully automated operations. The long-term objective is the creation of self-evolving communication infrastructures capable of continuously acquiring knowledge, adapting their operational behavior, validating their own decisions, and incorporating emerging technologies without requiring fundamental architectural redesign. Achieving this vision will require the development of scientific principles that are only beginning to emerge today. Consequently, the next decade represents not the culmination of network automation research, but rather the beginning of a new scientific era in which intelligence becomes an intrinsic property of communication infrastructures themselves. The research challenges identified in this paper are intended to stimulate further discussion and inspire future investigations toward the realization of truly cognitive, trustworthy, and autonomous network and service management systems.

TABLE II
RESEARCH PRIORITIES FOR THE NEXT DECADE

Research Area	Priority	Expected Outcome
Trustworthy AI	Very High	Reliable autonomous decisions
Agentic AI	Very High	Distributed operational intelligence
Semantic Telemetry	High	Context-aware observability
Digital Twins	High	Predictive network management
Large Network Models	High	Networking-specific reasoning
Knowledge Graphs	Medium	Semantic interoperability
Formal Verification	High	Safe AI-generated actions
Sustainable AI	Medium	Energy-efficient autonomous management

APPENDIX A

DISCLOSURE STATEMENT

This manuscript was generated using OpenAI ChatGPT (GPT-5.5) following the rules of the CNSM 2026 Agentic AI Researcher Track. The human research director defined the topic and constraints while the AI generated the technical content.

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